

The Utility Problem

As energy is consumed daily, the utility experiences losses, which are classified as technical or non-technical. The total losses that a utility experiences are quite simple to determine: subtract the energy billed from the energy supplied, or input into the substation depending on the reach of the utility. However, differentiating between technical losses and nontechnical losses and identifying the location are much more difficult tasks, requiring analytics to be done efficiently. Technical losses, while they can be improved upon by updating and maintaining grid components to increase efficiency, are inherent in the grid and are often of much smaller magnitude. However, non-technical losses, namely energy theft and failures (accidental or intended) of energy sensors, represent a much larger loss for the utility and can be eradicated through analytics.

Non-technical losses, as mentioned above, may occur for multiple reasons. Some examples as follows:

- An individual tampers with a power supply line to bypass the energy meter;
- An individual manipulates the energy meter to provide readings of less or no energy,
- An individual accidentally causes the energy meter to provide inaccurate readings;
- Or there is technical malfunction of the energy meter.

Thus, it is in the utility's interest to not only identify these losses but also eradicate them. This effort is greatly eased with smart meter data (if available) as utilities gain valuable insights through the interval readings provided, enabling more in-depth load profiles. Without data, this effort is very manual and inefficient, but with data, the utility is able to identify suspicious patterns in the energy consumption profiles and investigate them further to identify these nontechnical losses. This will then result in financial savings for the utility.

The Data Solutions and Methods

For utilities to be able to identify suspicious patterns which would indicate non-technical losses, the following datasets need to be available:

1. GIS/Connectivity:
 - Provides a topology model of the grid, allowing for analysis of power distribution from the substation to a customer meter,
2. Consumer Information
 - Consumer information from Client Information System (CIS) provides additional context into the analyzed data in the form of knowledge about:
 - i. Type of consumer: residential, commercial, industrial
 - ii. Tariff,
 - iii. Type of heating
 - iv. Presence of behind the meter DER,
 - v. etc.
3. AMI/AMR/Billing

- Provides data to indicate the energy billed for consumption for a particular consumer.

The Awesense Energy Transition Platform can ingest and cleanse all mentioned data types as well as synchronize the GIS/Connectivity with the time-series data, accelerating this type of analysis. The automated ingestion of data reduces the time to value for use cases as the Awesense data engine is able to complete this process much quicker than other industry tools. Along with ingestion, the Awesense data engine also performs the cleansing of the data, enabling utilities to get accurate results from their analyses while removing the need to manually cleanse the data in a tedious and time-consuming manner. Finally, the Awesense data engine also synchronizes the GIS/Connectivity data with the time-series data, allowing for more comprehensive and efficient analytics of the grid.

After the data has been processed by the Awesense data engine, analysis can be performed to determine suspicious patterns of consumption. With Awesense's comprehensive Energy Data Model (EDM), a data professional can focus on the desired segment/location of the grid where she or he would like to analyze the consumption patterns. Once the desired set of grid elements for the analysis has been retrieved, the analyst can then employ Awesense's predefined function, `ts_data_source_select ( )`, to gather all the time-series data for these specified grid elements.

At this point, the most difficult part of the analysis has been completed and the data needs to be analyzed for any outliers or patterns. This is a flexible process and can be done in countless ways, thus a specific process will not be identified. An analyst may choose to look at historical data and identify any significant changes in energy usage; an analyst may look at similar customers and compare the energy consumption profiles; an analyst may utilize a specialized algorithm to determine if the energy consumption profile is non-typical; or an analyst may choose to employ a combination of the approaches mentioned or approaches not mentioned as this is not an exhaustive list.

This type of analysis can also be completed through Awesense's frontend solution, True Grid Intelligence (TGI).

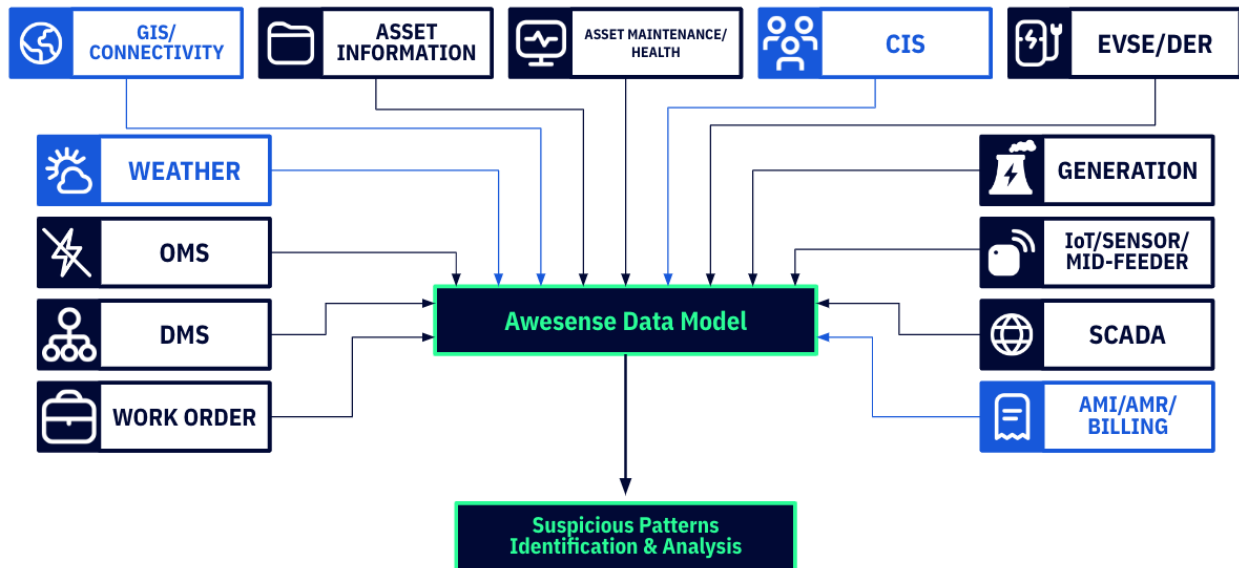
Suspicious Patterns Identification & Analysis

[UC30-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.